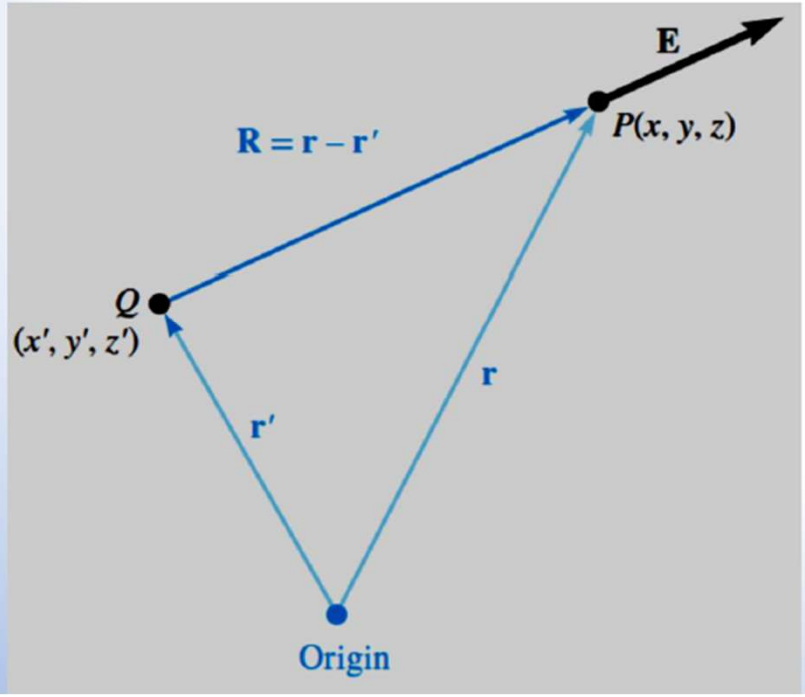
Electrostatics – Magneto-statics

- Static fields
- Time varying fields – constitutes waves
- Recap

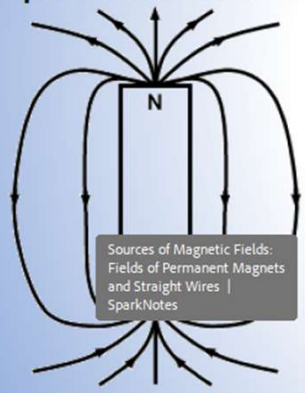


$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$

Magnetostatic- Biot Savart Law

The magnetic field produced by a differential dc element in free space

- Source of steady magnetic field – permanent magnet



- Magnetic filed also produced by differential DC element in free space

Dr.

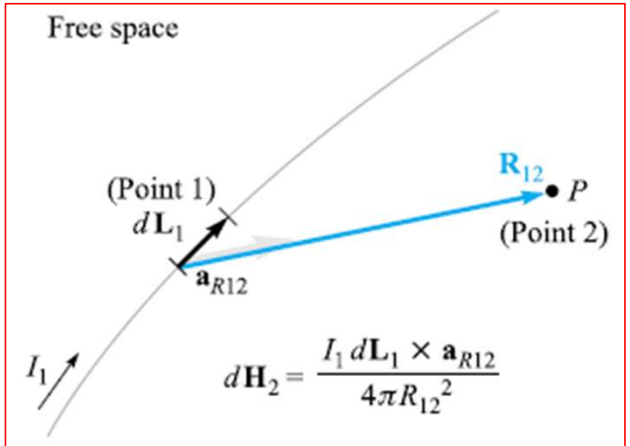
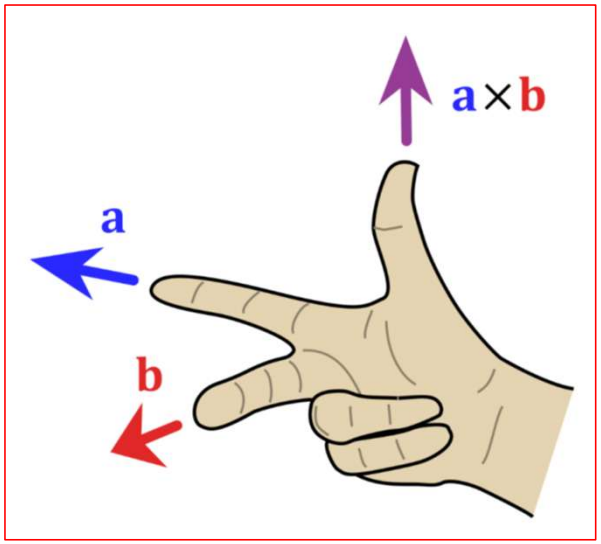


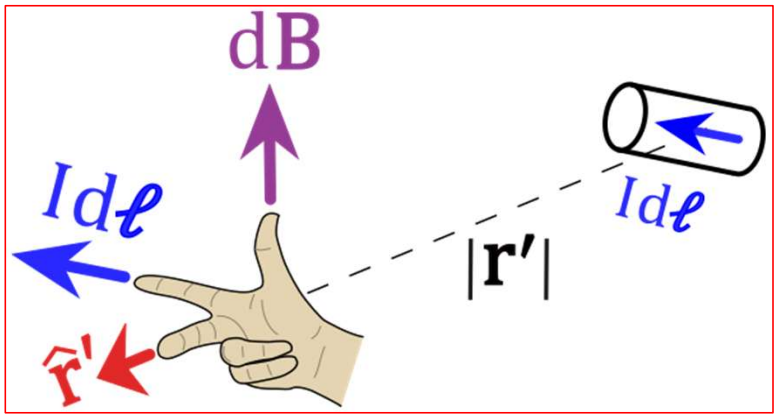
Figure 7.1 The law of Biot-Savart expresses the magnetic field intensity $d\mathbf{H}_2$ produced by a differential current element $I_1 d\mathbf{L}_1$. The direction of $d\mathbf{H}_2$ is into the page.

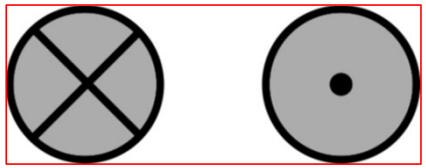
$$d\mathbf{H} = \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{Id\mathbf{L} \times \mathbf{R}}{4\pi R^3}$$

$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$

Understand Direction of Magnetic Field





Going into the paper  Coming out of paper

Magnetostatic- Biot Savart Law

The integral form of the Biot-Savart law can be verified experimentally

$$\mathbf{H} = \oint \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

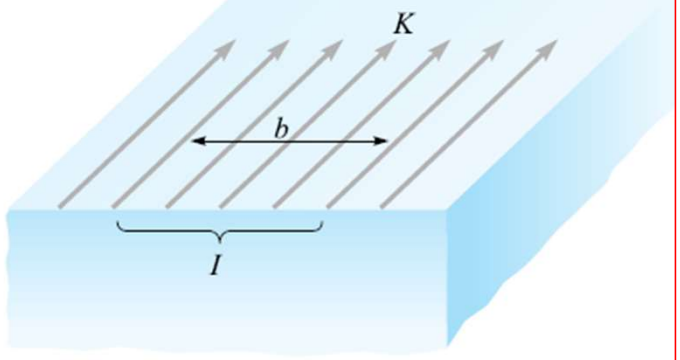


Figure 7.2 The total current I within a transverse width b, in which there is a *uniform* surface current density K, is Kb.

$$I\,d\mathbf{L} = \mathbf{K}\,dS = \mathbf{J}\,dv$$

$$\mathbf{H} = \int_S \frac{\mathbf{K} \times \mathbf{a}_R dS}{4\pi R^2}$$

$$\mathbf{H} = \int_{\text{vol}} \frac{\mathbf{J} \times \mathbf{a}_R dv}{4\pi R^2}$$

$$\mathbf{E} = \frac{Q_1}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$	$$\mathbf{H} = \oint \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$
ρl dl = ρs ds = ρv dv	IdL = KdS = JdV
$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho$$	$$\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi$$

Magnetic Field Due to Infinite Current Carrying Conductor

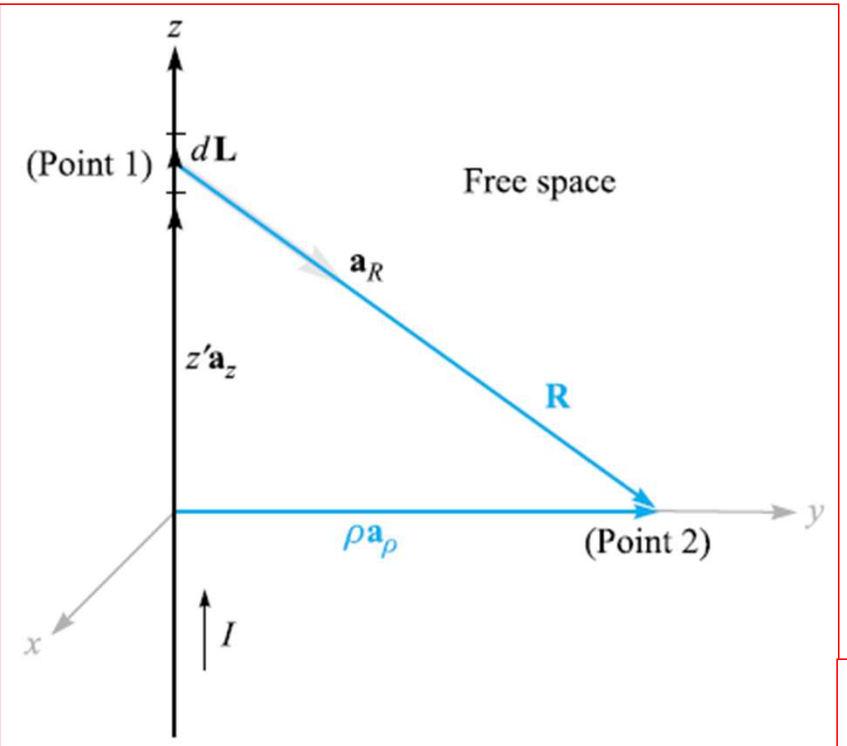


Figure 7.3 An infinitely long straight filament carrying a direct current I. The field at point 2 is $\mathbf{H} = (I/2\pi\rho)\mathbf{a}_\phi$.

$$\mathbf{R}_{12} = \mathbf{r} - \mathbf{r}' = \rho\mathbf{a}_\rho - z'\mathbf{a}_z$$

$$\mathbf{a}_{R12} = \frac{\rho\mathbf{a}_\rho - z'\mathbf{a}_z}{\sqrt{\rho^2 + z'^2}}$$

$$d\mathbf{H} = \frac{Id\mathbf{L} \times \mathbf{a}_R}{4\pi R^2} = \frac{Id\mathbf{L} \times \mathbf{R}}{4\pi R^3}$$

$$d\mathbf{H}_2 = \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{H}_2 = \int_{-\infty}^{\infty} \frac{I\,dz'\mathbf{a}_z \times (\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi(\rho^2 + z'^2)^{3/2}}$$

$$= \frac{I}{4\pi}\int_{-\infty}^{\infty} \frac{\rho dz'\mathbf{a}_\phi}{(\rho^2 + z'^2)^{3/2}}$$

$$\mathbf{H}_2 = \frac{I\rho\mathbf{a}_\phi}{4\pi}\int_{-\infty}^{\infty} \frac{dz'}{(\rho^2 + z'^2)^{3/2}}$$

$$= \frac{I\rho\mathbf{a}_\phi}{4\pi}\left.\frac{z'}{\rho^2\sqrt{\rho^2 + z'^2}}\right|_{-\infty}^{\infty}$$

$$\mathbf{H}_2 = \frac{I}{2\pi\rho}\mathbf{a}_\phi$$

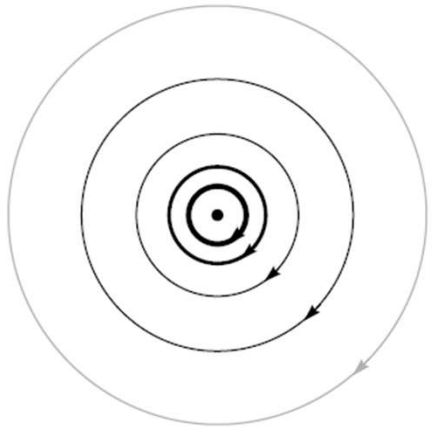


Figure 7.4 The streamlines of the magnetic field intensity about an infinitely long straight filament carrying a direct current I. The direction of I is into the page.

$$I = \int \frac{1}{(x^2 + a^2)^{3/2}}\, dx$$

Let's simplify it by making a substitution...

$$x = a \tan t$$

Taking derivative of both sides...

$$dx = a \sec^2 t \; dt$$

$$I = \int \frac{a \sec^2 t}{(a^2 \tan^2 t + a^2)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{(a^2 (\tan^2 t + 1))^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 (\tan^2 t + 1)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 (\sec^2 t)^{3/2}}\, dt$$

$$I = \int \frac{a \sec^2 t}{a^3 \sec^3 t}\, dt$$

$$I = \int \frac{1}{a^2 \sec t}\, dt$$

$$I = \int \frac{\cos t}{a^2}\, dt$$

$$I = \frac{1}{a^2} \int \cos t \, dt$$

$$I = \frac{\sin t}{a^2} + C$$

We had initially substituted $x = a \tan t$

That means $\tan t = \frac{x}{a}$

So, we have $\sin t = \frac{x}{\sqrt{a^2 + x^2}}$

$$I = \frac{\frac{x}{\sqrt{a^2 + x^2}}}{a^2} + \text{I}$$

$$I = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

$$I = \int \frac{1}{(x^2 + a^2)^{3/2}}\, dx \quad = \frac{x}{a^2 \sqrt{a^2 + x^2}}$$

[You can take it as a standard result/formula]

In free space, we define the *magnetic flux density* **B** as

$$\mathbf{B} = \mu_0 \mathbf{H}$$ (free space only)

webers per square meter (Wb/m^2) or tesla (T).

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

The name given to μ_0 is the *permeability* of free space.

Problem 7.1 : Find the incremental field $\Delta \overline{H}_2$ at P_2 caused by a source at P_1 of $I_1 \Delta \overline{L}_1 =$: (a) $2\pi \bar{a}_z \mu$ A · m, given P_1 (4, 0, 0) and P_2 (0, 3, 0); (b) $2\pi \bar{a}_z \mu$ A · m, given P_1 (4, −2, 3) and P_2 (0, 3, 0); (c) $2\pi (0.6\bar{a}_x - 0.8\bar{a}_y) \mu$ A · m, given P_1 (4, −2, 3) and P_2 (1, 3, 2).

$$\Delta \mathbf{H} = \frac{I \Delta \mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

(a) $d\bar{H}_2 = \dfrac{I\,\overline{dL} \times \bar{a}_{12}}{4\pi d^2}$

Point 2 is that point at which magnetic field intensity is desired = (0, 3, 0)

Point1 is that point at which differential current element is located= (4, 0, 0)

$\bar{R}_{12} = (0 - 4)\,\bar{a}_x + (3 - 0)\,\bar{a}_y = -4\,\bar{a}_x + 3\,\bar{a}_y$

$d = |\bar{R}_{12}| = \sqrt{(-4)^2 + (3)^2} = 5$

$$\therefore\ d\bar{H}_2 = \frac{(2\pi 10^{-6}\,\bar{a}_z)}{4\pi (5)^2} \times \left[\frac{-4\,\bar{a}_x + 3\,\bar{a}_y}{5}\right] = -12\,\bar{a}_x - 16\,\bar{a}_y \text{ nA/m}$$

(b) $\bar{R}_{12} = (0 - 4)\,\bar{a}_x + (3 + 2)\,\bar{a}_y + (0 - 3)\,\bar{a}_z$

$= -4\,\bar{a}_x + 5\,\bar{a}_y - .3\,\bar{a}_z$

$$d = |\bar{R}_{12}| = \sqrt{(-4)^2 + (5)^2 + (-3)^2} = \sqrt{50} = 7.071$$

$$\therefore d\bar{H}_2 = \frac{(2\pi\, 10^{-6}\,\bar{a}_z)}{4\pi\,(7.071)^2} \times \left[\frac{-4\,\bar{a}_x + 5\,\bar{a}_y - 3\,\bar{a}_z}{7.071}\right]$$

$= -7.071\,\bar{a}_x - 5.657\,\bar{a}_y$ n A/m

(c) $\bar{R}_{12} = (1 - 4)\,\bar{a}_x + (3 + 2)\,\bar{a}_y + (2 - 3)\,\bar{a}_z$

$= -3\,\bar{a}_x + 5\,\bar{a}_y - \bar{a}_z$

$$d = |\bar{R}_{12}| = \sqrt{(-3)^2 + (5)^2 + (-1)^2} = \sqrt{35} = 5.916$$

$$\therefore d\bar{H}_2 = \left[\frac{2\pi\,(0.6\,\bar{a}_x - 0.8\,\bar{a}_y)\,10^{-6}}{4\pi\,(5.916)^2}\right] \times \left[\frac{-3\,\bar{a}_x + 5\,\bar{a}_y - \bar{a}_z}{5.916}\right]$$

$$= 1.932\,\bar{a}_x + 1.44\,\bar{a}_y + 1.449\,\bar{a}_z \quad \text{nA/m}$$

Magnetic Field Due to Finite-Length Current Filament

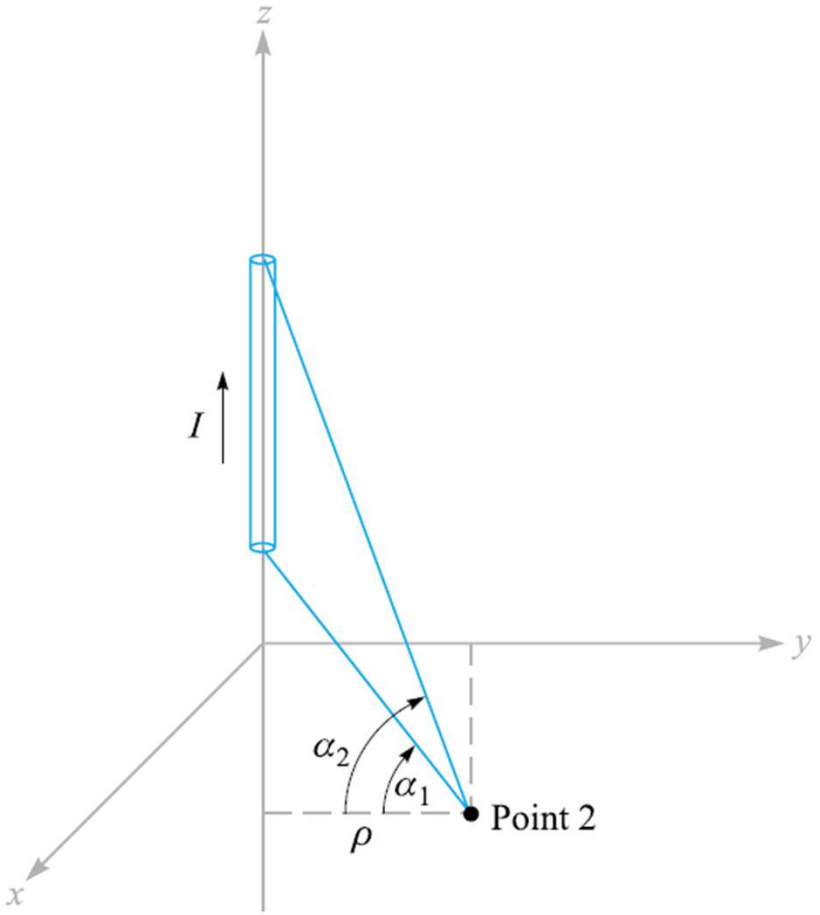


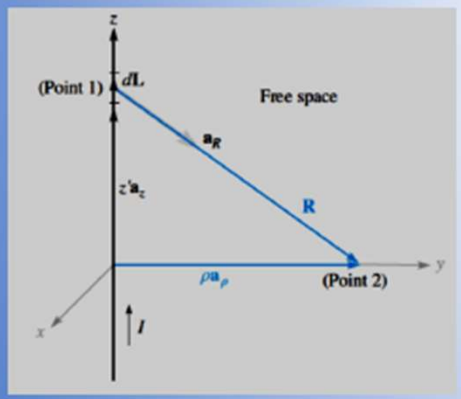
Figure 7.5 The magnetic field intensity caused by a finite-length current filament on the z axis is $(I/4\pi\rho)(\sin\alpha_2 - \sin\alpha_1)\mathbf{a}_\phi$.

$$\mathbf{H} = \frac{I}{4\pi\rho}(\sin\alpha_2 - \sin\alpha_1)\mathbf{a}_\phi$$

$$\mathbf{H} = \int_{z_1}^{z_2} \frac{I d\mathbf{L} \times \mathbf{a}_R}{4\pi R^2}$$

$$= \int_{\rho \tan \alpha_1}^{\rho \tan \alpha_2} \frac{I dz \mathbf{a}_z \times (\rho \mathbf{a}_\rho - z \mathbf{a}_z)}{4\pi(\rho^2 + z^2)^{3/2}}$$

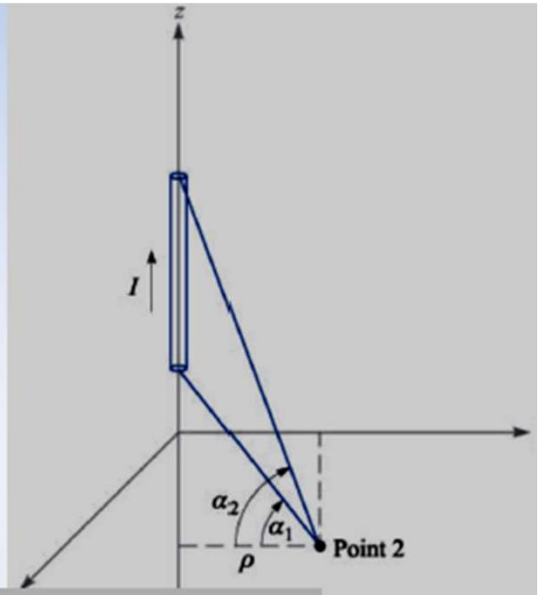
$$= \int_{\rho \tan \alpha_1}^{\rho \tan \alpha_2} \frac{I \rho \mathbf{a}_\phi \, dz}{4\pi(\rho^2 + z^2)^{3/2}}$$



$$= \frac{I z \mathbf{a}_\phi}{4\pi\rho\sqrt{\rho^2 + z^2}} \Bigg|_{\rho \tan \alpha_1}^{\rho \tan \alpha_2}$$

$$= \frac{I}{4\pi\rho} \left[\frac{\tan \alpha_2}{\sqrt{1 + \tan^2 \alpha_2}} - \frac{\tan \alpha_1}{\sqrt{1 + \tan^2 \alpha_1}} \right] \mathbf{a}_\phi$$

$$= \frac{I}{4\pi\rho} (\sin \alpha_2 - \sin \alpha_1) \mathbf{a}_\phi$$



$$I = \int \frac{1}{(x^2 + a^2)^{3/2}} dx = \frac{x}{a^2\sqrt{a^2 + x^2}}$$

← Formula

Find H at the center of a square current loop of side L.

Choose a Cartesian coordinate system such that the loop is located as shown in Fig. 10-10. By symmetry, each half-side contributes the same amount to H at the center. For the half-side $0 \le x \le L/2, y = -L/2$, the Biot-Savart law gives for the field at the origin

$$d\mathbf{H} = \frac{(I\,dx\,\mathbf{a}_x)\times[-x\mathbf{a}_x + (L/2)\mathbf{a}_y]}{4\pi[x^2 + (L/2^2)]^{3/2}} = \frac{I\,dx\,(L/2)\mathbf{a}_z}{4\pi[x^2 + (L/2^2)]^{3/2}}$$

Therefore, the total field at the origin is

$$\mathbf{H} = 8\int_0^{L/2} \frac{I\,dx\,(L/2)\mathbf{a}_z}{4\pi[x^2 + (L/2^3)]^{3/2}}$$

$$= \frac{2\sqrt{2}\,I}{\pi L}\mathbf{a}_z$$

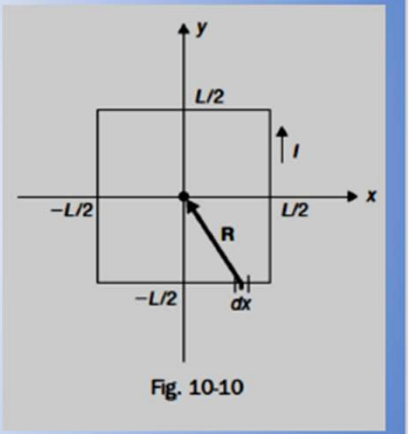


Fig. 10-10

Note: Easier way to solve it –use the formula

$$\frac{I}{4\pi\rho}(\sin\alpha_2 - \sin\alpha_1)\mathbf{a}_\phi$$

And $\alpha = 45^0$

Try this solved example from the text book

EXAMPLE 7.1

As a numerical example illustrating the use of (9), we determine **H** at $P_2(0.4, 0.3, 0)$ in the field of an 8-ampere filamentary current. The current is directed inward from infinity to the origin on the positive x axis, and then outward to infinity along the y axis. This arrangement is shown in Figure 7.6.

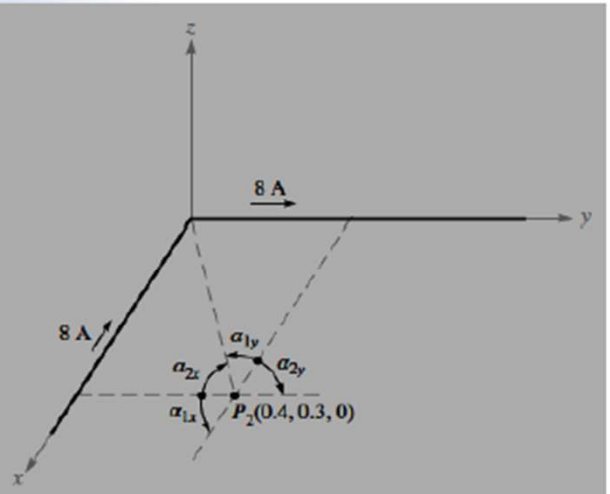


Figure 7.6 The individual fields of two semi-infinite current segments are found by (9) and added to obtain $\mathbf{H}_2$ at P_2.

$$\mathbf{H}_2 = \mathbf{H}_{2(x)} + \mathbf{H}_{2(y)} = -\frac{20}{\pi}\mathbf{a}_z = -6.37\mathbf{a}_z \text{ A/m}$$

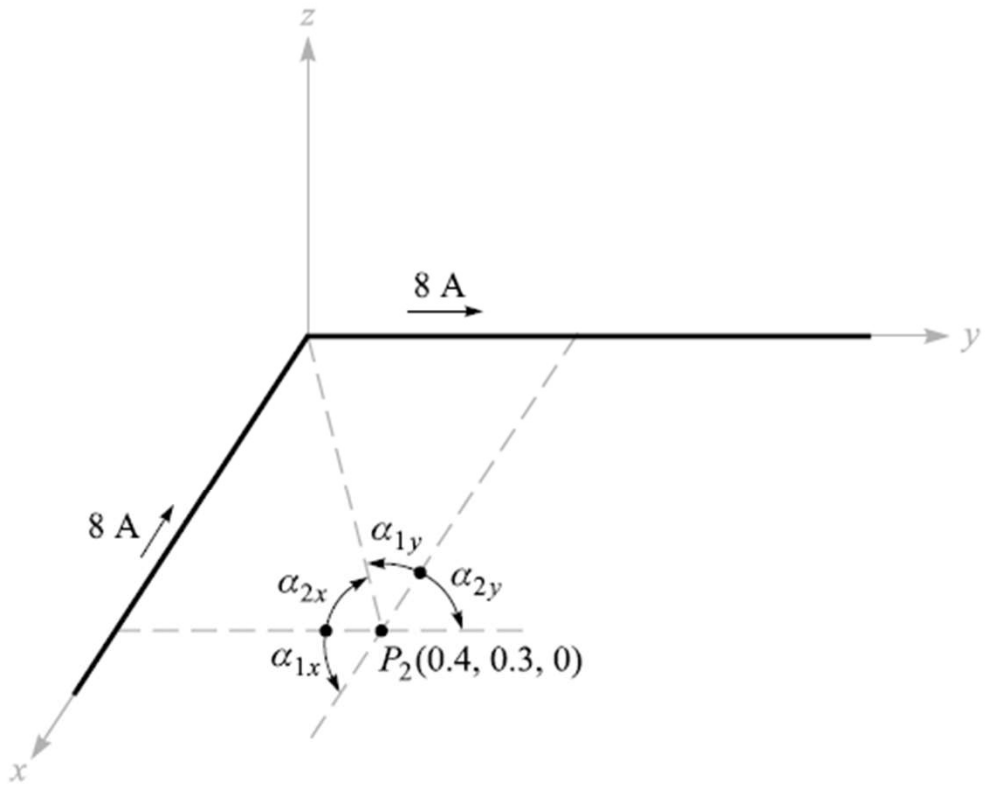


Figure 7.6 The individual fields of two semi-infinite current segments are found by (9) and added to obtain $\mathbf{H}_2$ at P_2.

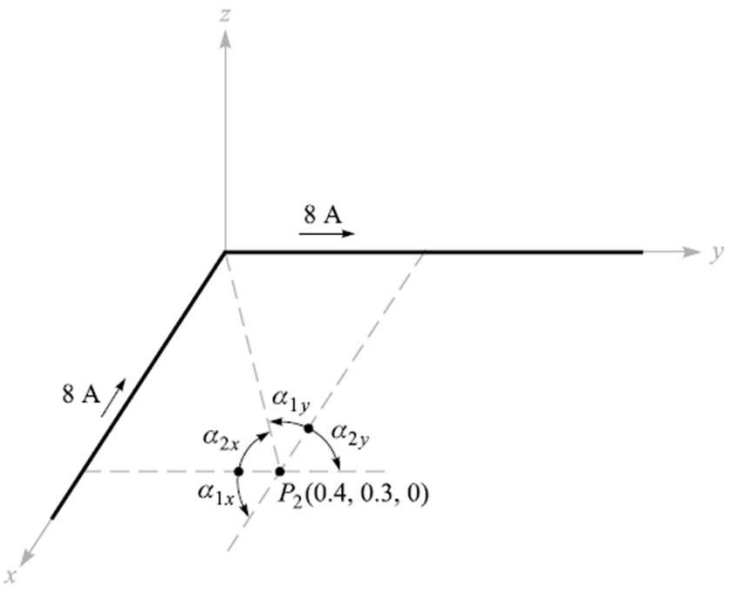


Figure 7.6 The individual fields of two semi-infinite current segments are found by (9) and added to obtain $\mathbf{H}_2$ at P_2.

Solution. We first consider the semi-infinite current on the x axis, identifying the two angles, $\alpha_{1x} = -90°$ and $\alpha_{2x} = \tan^{-1}(0.4/0.3) = 53.1°$. The radial distance ρ is measured from the x axis, and we have $\rho_x = 0.3$. Thus, this contribution to $\mathbf{H}_2$ is

$$\mathbf{H}_{2(x)} = \frac{8}{4\pi(0.3)}(\sin 53.1° + 1)\mathbf{a}_\phi = \frac{2}{0.3\pi}(1.8)\mathbf{a}_\phi = \frac{12}{\pi}\mathbf{a}_\phi$$

The unit vector $\mathbf{a}_\phi$ must also be referred to the x axis. We see that it becomes $-\mathbf{a}_z$. Therefore,

$$\mathbf{H}_{2(x)} = -\frac{12}{\pi}\mathbf{a}_z \text{ A/m}$$

For the current on the y axis, we have $\alpha_{1y} = -\tan^{-1}(0.3/0.4) = -36.9°$, $\alpha_{2y} = 90°$, and $\rho_y = 0.4$. It follows that

$$\mathbf{H}_{2(y)} = \frac{8}{4\pi(0.4)}(1 + \sin 36.9°)(-\mathbf{a}_z) = -\frac{8}{\pi}\mathbf{a}_z \text{ A/m}$$

Ampère's circuital law states that the line integral of **H** about any *closed* path is exactly equal to the direct current enclosed by that path,

$$\oint \mathbf{H} \cdot d\mathbf{L} = I \tag{10}$$

We define positive current as flowing in the direction of advance of a right-handed screw turned in the direction in which the closed path is traversed.

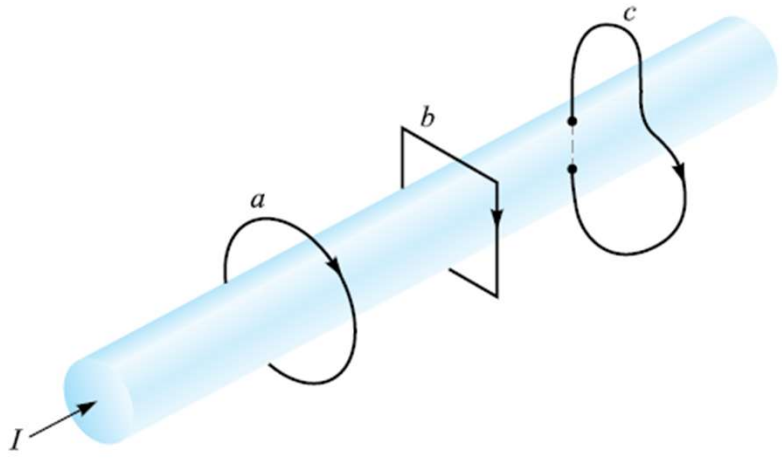


Figure 7.7 A conductor has a total current *I*. The line integral of H about the closed paths *a* and *b* is equal to *I*, and the integral around path *c* is less than *I*, since the entire current is not enclosed by the path.

- Find H due to infinitely long current carrying conductor

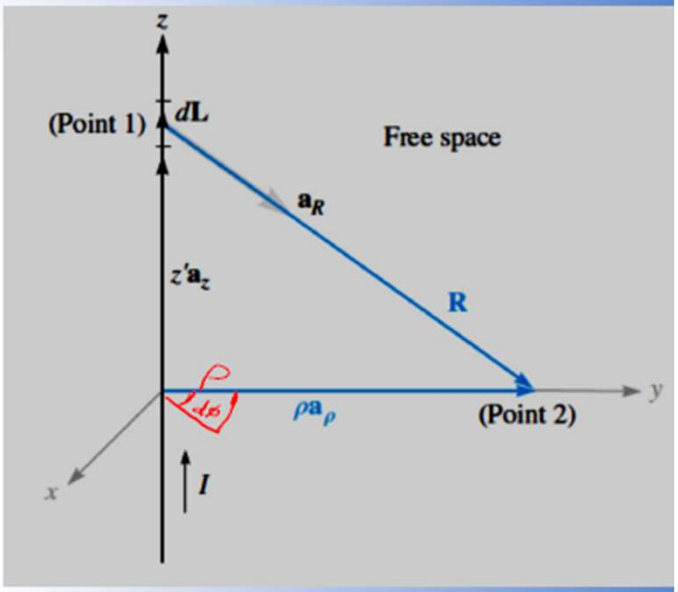
Direction of *d*H is perpendicular to the plane containing *d*L and R and therefore is in the direction of $\mathbf{a}_\phi$

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{2\pi} H_\phi \rho d\phi$$

$$= H_\phi \rho \int_0^{2\pi} d\phi$$

$$= H_\phi 2\pi\rho = I$$

$$H_\phi = \frac{I}{2\pi\rho}$$



Infinitely long coaxial transmission line carrying a uniformly distributed total current I in the center conductor and −I in the outer conductor

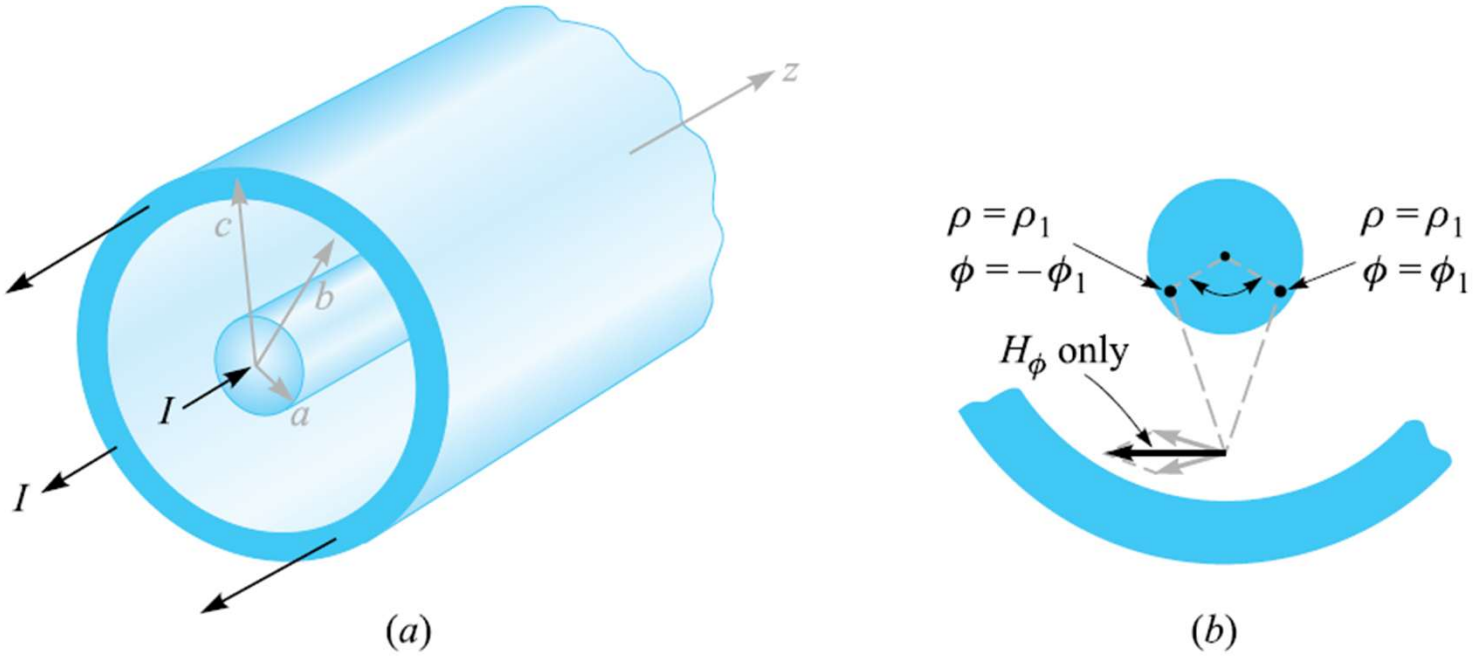
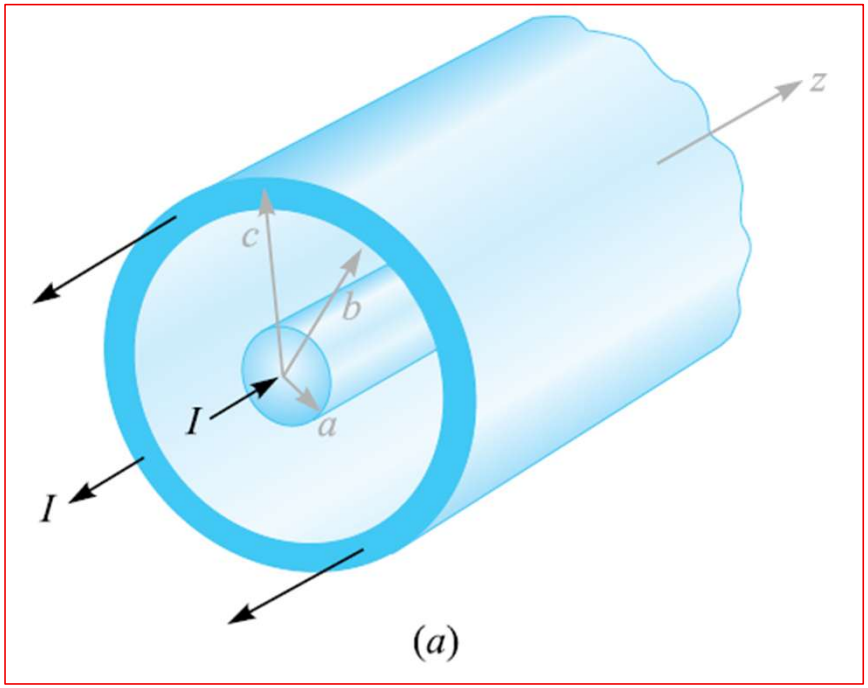
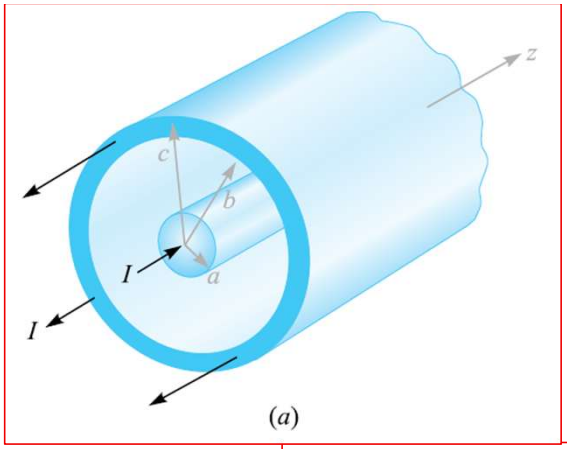


Figure 7.8 (*a*) Cross section of a coaxial cable carrying a uniformly distributed current I in the inner conductor and $-I$ in the outer conductor. The magnetic field at any point is most easily determined by applying Ampère's circuital law about a circular path. (*b*) Current filaments at $\rho = \rho_1, \phi = \pm\phi_1$, produces H_ρ components which cancel. For the total field, $\mathbf{H} = H_\phi \mathbf{a}_\phi$.



$$H_\phi = \frac{I}{2\pi\rho} \quad (a < \rho < b)$$



(a)

If we choose ρ smaller than the radius of the inner conductor, the current enclosed is

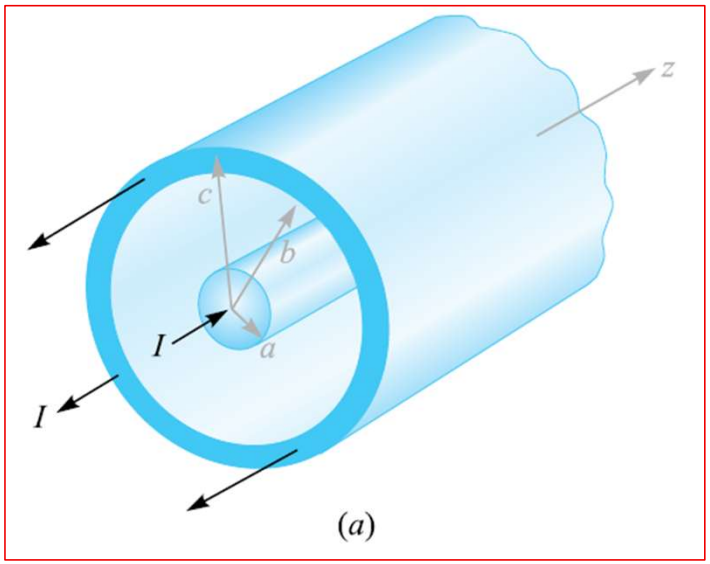
$$I_{\text{encl}} = I\frac{\rho^2}{a^2}$$

and

$$2\pi\rho H_\phi = I\frac{\rho^2}{a^2}$$

or

$$H_\phi = \frac{I\rho}{2\pi a^2} \quad (\rho < a)$$



(a)

if the path lies within the outer conductor, we have

$$2\pi\rho H_\phi = I - I\left(\frac{\rho^2 - b^2}{c^2 - b^2}\right)$$

$$H_\phi = \frac{I}{2\pi\rho}\frac{c^2 - \rho^2}{c^2 - b^2} \quad (b < \rho < c)$$

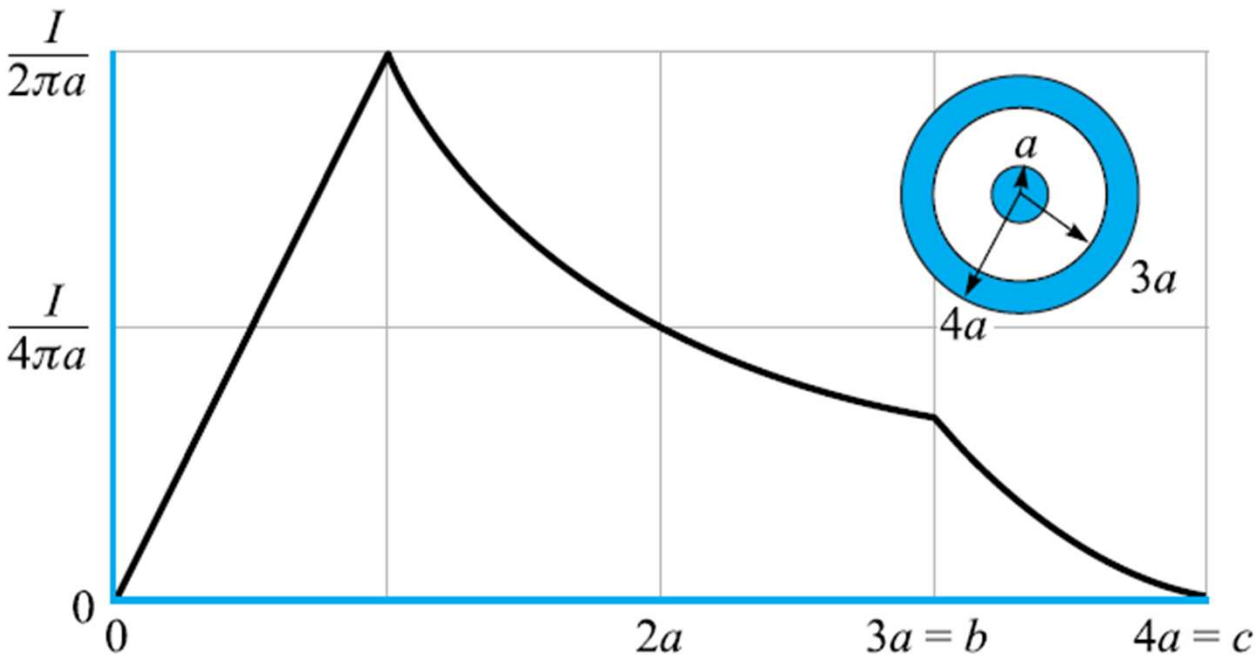


Figure 7.9 The magnetic field intensity as a function of radius in an infinitely long coaxial transmission line with the dimensions shown.

$$H_\phi = \frac{I\rho}{2\pi a^2} \quad (\rho < a)$$

$$\boldsymbol{H_\phi = \frac{I}{2\pi\rho} \quad (a < \rho < b)}$$

$$H_\phi = \frac{I}{2\pi\rho}\frac{c^2 - \rho^2}{c^2 - b^2} \quad (b < \rho < c)$$

Find **H** on the axis of a circular current loop of radius a. Specialize the result to the center of the loop.

For the point shown in Fig. 10-15,

$$\mathbf{R} = -a\mathbf{a}_r + h\mathbf{a}_z$$

$$d\mathbf{H} = \frac{(Ia\,d\phi\,\mathbf{a}_\phi)\times(-a\mathbf{a}_r + h\mathbf{a}_z)}{4\pi(a^2+h^2)^{3/2}} = \frac{(Ia\,d\phi)(a\mathbf{a}_z + h\mathbf{a}_r)}{4\pi(a^2+h^2)^{3/2}}$$

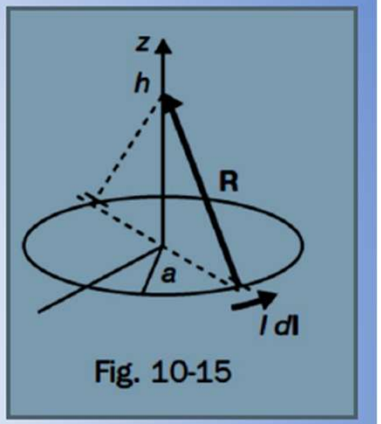


Fig. 10-15

$$\mathbf{H} = \int_0^{2\pi} \frac{Ia^2\,d\phi}{4\pi(a^2+h^2)^{3/2}}\mathbf{a}_z$$

$$= \frac{Ia^2}{2(a^2+h^2)^{3/2}}\mathbf{a}_z$$

At $h = 0$, $\mathbf{H} = (I/2a)\mathbf{a}_z$.

H due to uniform surface current density Ky ay

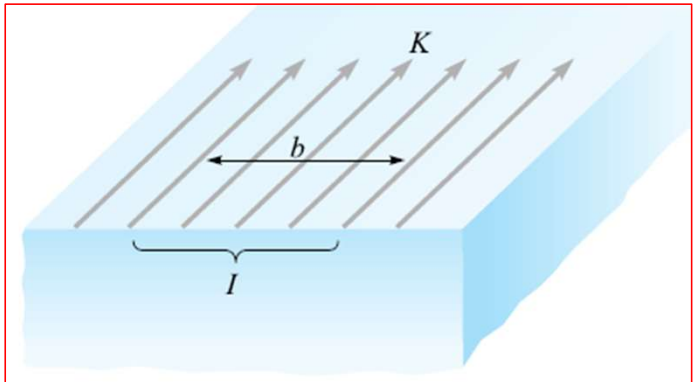


Figure 7.2 The total current I within a transverse width b, in which there is a *uniform* surface current density K, is Kb.

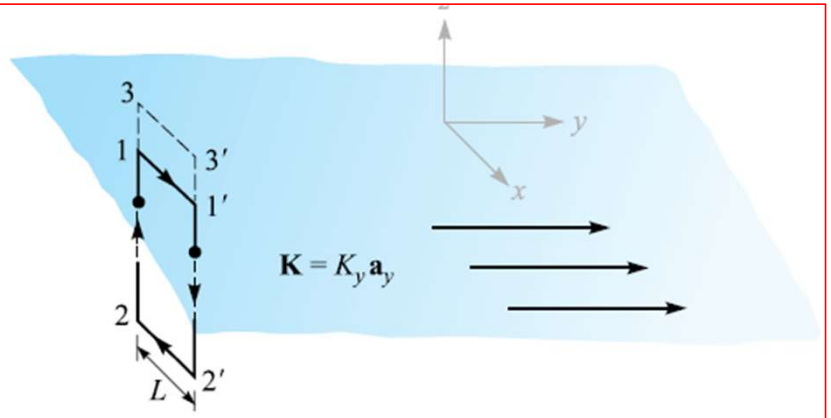


Figure 7.10 A uniform sheet of surface current $\mathbf{K} = K_y\mathbf{a}_y$ in the $z = 0$ plane. $\mathbf{H}$ may be found by applying Ampère's circuital law about the paths 1-1′-2′-2-1 and 3-3′-2′-2-3.

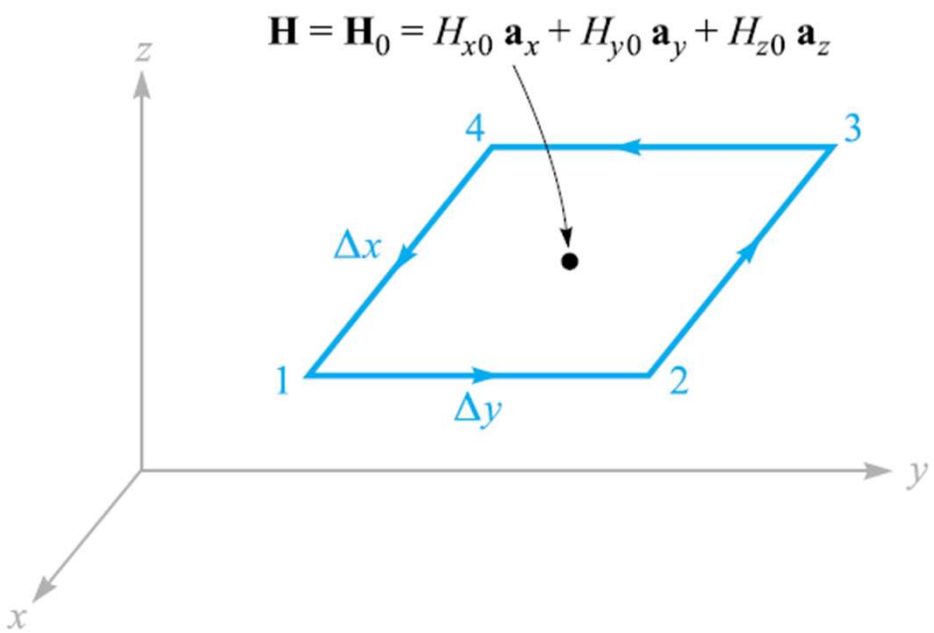


Figure 7.13 An incremental closed path in rectangular coordinates is selected for the application of Ampère's circuital law to determine the spatial rate of change of H.

Curl

- Note: Gauss law-differential volume element – Divergence
- Here, Amperes law - differential surface element – Curl

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

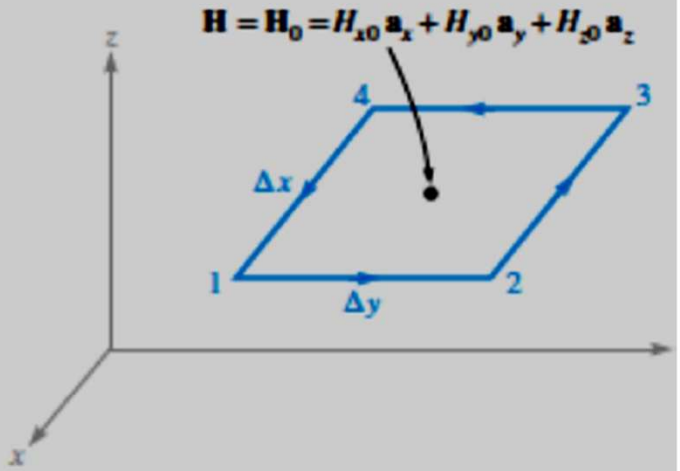


Figure 7.13 An incremental closed path in rectangular coordinates is selected for the application of Ampère's circuital law to determine the spatial rate of change of **H**.

H at the *center* of this small rectangle

$$\mathbf{H}_0 = H_{x0}\mathbf{a}_x + H_{y0}\mathbf{a}_y + H_{z0}\mathbf{a}_z$$

The closed line integral of **H** about this path is then approximately the sum of the four values of $\mathbf{H} \cdot \Delta\mathbf{L}$ on each side.

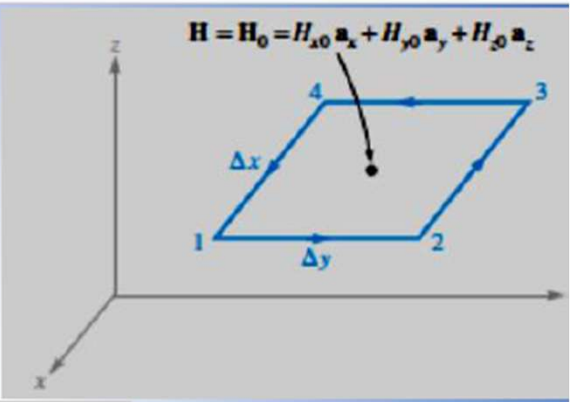
We choose the direction of traverse as 1-2-3-4-1, which corresponds to a current in the $\mathbf{a}_z$ direction, and the first contribution is therefore

$$(\mathbf{H} \cdot \Delta\mathbf{L})_{1-2} = H_{y,1-2}\Delta y$$

$$(\mathbf{H}\cdot\Delta\mathbf{L})_{1-2}=H_{y,1-2}\Delta y$$

$$H_{y,1-2}\doteq H_{y0}+\frac{\partial H_y}{\partial x}\left(\frac{1}{2}\Delta x\right)$$

$$(H\cdot\Delta L)_{1-2}\doteq\left(H_{y0}+\frac{1}{2}\frac{\partial H_y}{\partial x}\Delta x\right)\Delta y$$



Similarly $$(\mathbf{H}\cdot\Delta\mathbf{L})_{2-3}\doteq H_{x,2-3}(-\Delta x)\doteq-\left(H_{x0}+\frac{1}{2}\frac{\partial H_x}{\partial y}\Delta y\right)\Delta x$$

Continuing for the remaining two segments and adding the results,

$$\oint\mathbf{H}\cdot d\mathbf{L}\doteq\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\Delta x\Delta y$$

current density J, the enclosed current is then $\Delta I\doteq J_z\Delta x\Delta y$, and

$$\oint\mathbf{H}\cdot d\mathbf{L}\doteq\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\Delta x\Delta y\doteq J_z\Delta x\Delta y$$

or

$$\frac{\oint\mathbf{H}\cdot d\mathbf{L}}{\Delta x\Delta y}\doteq\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\doteq J_z$$

- As we cause the closed path to shrink

$$\lim_{\Delta x,\Delta y\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta x\Delta y}=\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}=J_z$$

- Analogous processes lead to expressions for the *x* and *y* components of the current density,

$$\lim_{\Delta y,\Delta z\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta y\Delta z}=\frac{\partial H_z}{\partial y}-\frac{\partial H_y}{\partial z}=J_x$$

$$\lim_{\Delta z,\Delta x\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta z\Delta x}=\frac{\partial H_x}{\partial z}-\frac{\partial H_z}{\partial x}=J_y$$

- Given

$$(\text{curl }\mathbf{H})_N=\lim_{\Delta S_N\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta S_N}$$

$$\text{curl }\mathbf{H}=\left(\frac{\partial H_z}{\partial y}-\frac{\partial H_y}{\partial z}\right)\mathbf{a}_x+\left(\frac{\partial H_x}{\partial z}-\frac{\partial H_z}{\partial x}\right)\mathbf{a}_y+\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\mathbf{a}_z$$

$$\text{curl }\mathbf{H}=\begin{vmatrix}\mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ H_x & H_y & H_z\end{vmatrix}$$

vector operator

$$\text{curl }\mathbf{H}=\nabla\times\mathbf{H}$$

Curl in cylindrical and spherical coordinates

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho}\frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z}\right)\mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right)\mathbf{a}_\phi + \left(\frac{1}{\rho}\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{1}{\rho}\frac{\partial H_\rho}{\partial \phi}\right)\mathbf{a}_z \quad \text{(cylindrical)}$$

$$\nabla \times \mathbf{H} = \frac{1}{r\sin\theta}\left(\frac{\partial (H_\phi \sin\theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi}\right)\mathbf{a}_r + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial H_r}{\partial \phi} - \frac{\partial (rH_\phi)}{\partial r}\right)\mathbf{a}_\theta + \frac{1}{r}\left(\frac{\partial (rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta}\right)\mathbf{a}_\phi \quad \text{(spherical)}$$

Curl Formula in Cylindrical

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Curl Formula in Spherical

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

As an example of the evaluation of curl **H** from the definition and of the evaluation of another line integral, suppose that $\mathbf{H} = 0.2z^2\mathbf{a}_x$ for $z > 0$, and $\mathbf{H} = 0$ elsewhere, as shown in Figure 7.15. Calculate $\oint \mathbf{H} \cdot d\mathbf{L}$ about a square path with side d, centered at $(0, 0, z_1)$ in the $y = 0$ plane where $z_1 > d/2$.

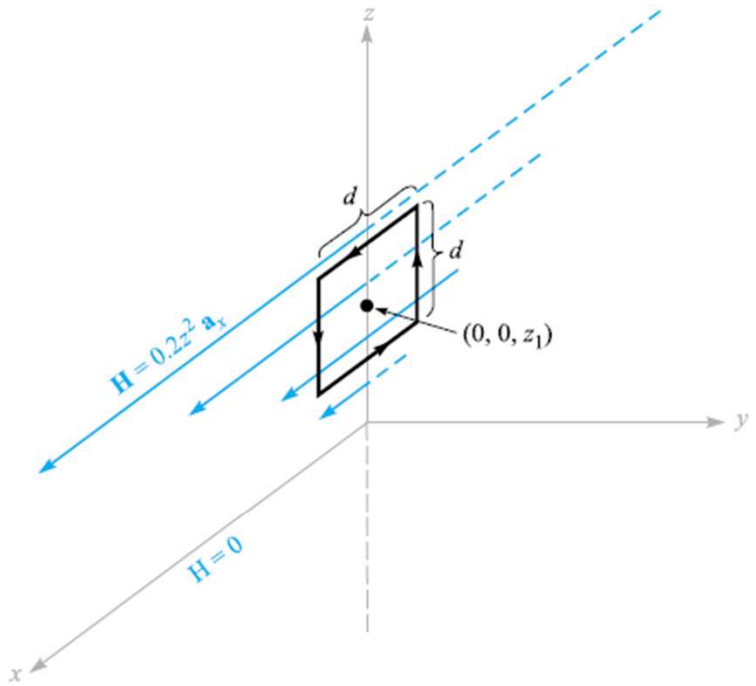


Figure 7.15 A square path of side d with its center on the z axis at $z = z_1$ is used to evaluate $\oint \mathbf{H} \cdot d\mathbf{L}$ and find curl **H**.

Solution. We evaluate the line integral of **H** along the four segments, beginning at the top:

$$\oint \mathbf{H} \cdot d\mathbf{L} = 0.2\left(z_1 + \tfrac{1}{2}d\right)^2 d + 0 - 0.2\left(z_1 - \tfrac{1}{2}d\right)^2 d + 0$$
$$= 0.4z_1 d^2$$

In the limit as the area approaches zero, we find

$$(\nabla \times \mathbf{H})_y = \lim_{d\to 0} \frac{\oint \mathbf{H} \cdot d\mathbf{L}}{d^2} = \lim_{d\to 0} \frac{0.4z_1 d^2}{d^2} = 0.4z_1$$

The other components are zero, so $\nabla \times \mathbf{H} = 0.4z_1\mathbf{a}_y$.

To evaluate the curl without trying to illustrate the definition or the evaluation of a line integral, we simply take the partial derivative indicated by (23):

$$\nabla \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 0.2z^2 & 0 & 0 \end{vmatrix} = \frac{\partial}{\partial z}(0.2z^2)\mathbf{a}_y = 0.4z\mathbf{a}_y$$

which checks with the preceding result when $z = z_1$.

Example 3.14

The magnetic field intensity in free space is given by $\mathbf{H} = 10\rho^2\,\mathbf{a}_\phi$ (A/m).

a. Find $\mathbf{J}$.

b. Integrate $\mathbf{J}$ over the circular surface, $\rho = 1$, all ϕ, and $z = 0$.

point form of Ampère's circuital law,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho}\frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z}\right)\mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right)\mathbf{a}_\phi + \left(\frac{1}{\rho}\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{1}{\rho}\frac{\partial H_\rho}{\partial \phi}\right)\mathbf{a}_z \quad \text{(cylindrical)}$$

Use the curl in cylindrical coordinates. For a vector $\mathbf{A} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z$,

$$(\nabla \times \mathbf{A})_\rho = \frac{1}{\rho}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z},$$
$$(\nabla \times \mathbf{A})_\phi = \frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho},$$
$$(\nabla \times \mathbf{A})_z = \frac{1}{\rho}\left[\frac{\partial(\rho A_\phi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi}\right].$$

Here $A_\rho = 0,\ A_\phi = 10\rho^2,\ A_z = 0$. So

- $(\nabla \times \mathbf{H})_\rho = \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} - \frac{\partial(10\rho^2)}{\partial z} = 0 - 0 = 0.$
- $(\nabla \times \mathbf{H})_\phi = \frac{\partial(0)}{\partial z} - \frac{\partial(0)}{\partial \rho} = 0 - 0 = 0.$
- $(\nabla \times \mathbf{H})_z = \frac{1}{\rho}\frac{\partial(\rho \cdot 10\rho^2)}{\partial \rho} = \frac{1}{\rho}\frac{\partial(10\rho^3)}{\partial \rho} = \frac{1}{\rho}(30\rho^2) = 30\rho.$

Thus

$$\boxed{\mathbf{J} = 30\rho\,\hat{a}_z \quad (\mathrm{A/m^2})}.$$

$$I = \int\int \mathbf{J} \cdot d\mathbf{S}$$
$$= \int_{\rho=0}^{1}\int_{\phi=0}^{2\pi} 30\rho\,\mathbf{a}_z \cdot \rho d\rho d\phi\,\mathbf{a}_z$$
$$= 30 \times 2\pi \int_{\rho=0}^{1} \rho^2 d\rho$$
$$= 30 \times 2\pi \times \frac{1}{3} = 20\pi \ (\mathrm{A})$$

point form of Ampère's circuital law,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

STOKES' THEOREM:

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$

Note: Similar to Divergence theorem which relates volume integral and surface integral. Here surface integral and line integrals are related.

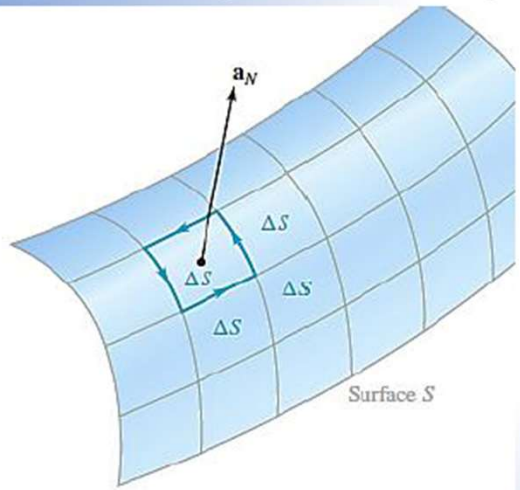


Figure 7.16 The sum of the closed line integrals about the perimeter of every ΔS is the same as the closed line integral about the perimeter of S because of cancellation on every interior path.

Proof: If we apply the definition of the curl to one of these incremental surfaces, then

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H})_N$$

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N$$

$$\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N \Delta S = (\nabla \times \mathbf{H}) \cdot \Delta \mathbf{S}$$

Every *interior* wall is covered once in each direction.
The only boundaries on which cancellation cannot occur form the outside boundary, the path enclosing *S*.
Therefore we have

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

In cylindrical coordinates, the magnetic field is given as $\mathbf{H} = [4\rho - 2\rho^2]\,\mathbf{a}_\phi$ (A/m), $0 \le \rho \le 1$.

a. Find the current density as a function of ρ within the cylinder.

b. Find the total current that passes through the surface $z = 0$ and $0 \le \rho \le 1$ (m) in the $\mathbf{a}_z$ direction.

c. Check the result of part (b) by another method. — Stoke's theorem

In cylindrical coordinates, the magnetic field is given as $\mathbf{H} = [4\rho - 2\rho^2]\,\mathbf{a}_\phi$ (A/m), $0 \le \rho \le 1$.

a. Find the current density as a function of ρ within the cylinder.

b. Find the total current that passes through the surface $z = 0$ and $0 \le \rho \le 1$ (m) in the $\mathbf{a}_z$ direction.

c. Check the result of part (b) by another method. — Stoke's theorem

$$\mathbf{J} = \nabla \times \mathbf{H}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ H_\rho & \rho H_\phi & H_z \end{vmatrix}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & 0 & 0 \\ 0 & \rho H_\phi & 0 \end{vmatrix}$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho H_\phi)\,\mathbf{a}_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left[\rho \times 4\rho - \rho \times 2\rho^2\right]\mathbf{a}_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left[4\rho^2 - 2\rho^3\right]\mathbf{a}_z$$

$$= \frac{1}{\rho}\left\{8\rho - 6\rho^2\right\}\mathbf{a}_z$$

$$= (8 - 6\rho)\,\mathbf{a}_z \ (\mathrm{A/m^2})$$

$$I = \int_S \mathbf{J}\cdot d\mathbf{S}$$

$$= \int_S J_z\,\mathbf{a}_z \cdot \rho\, d\rho\, d\phi\, \mathbf{a}_z$$

$$= \int_{\rho=0}^{1}\int_{\phi=0}^{2\pi} (8 - 6\rho)\rho\, d\rho\, d\phi$$

$$= 2\pi \int_{\rho=0}^{1} (8\rho - 6\rho^2)\, d\rho$$

$$= 2\pi \left\{\frac{8\rho^2}{2} - 6 \times \frac{\rho^3}{3}\right\}_{\rho=0}^{1}$$

$$= 2\pi \left\{4\rho^2 - 2\rho^3\right\}_0^1$$

$$= 2\pi \times 2 = 4\pi \ (\mathrm{A})$$

c) $I = \oint H \cdot dl$

$$= \int_{\phi=0}^{2\pi} (4\rho - 2\rho^2)\hat{a}_\phi \cdot \rho\, d\phi\, \hat{a}_\rho \Big]_{\rho=1}$$

$$= \int_0^{2\pi} (4-2)\cdot 1\, d\phi$$

$$= 2 \cdot 2\pi = 4\pi \ \mathrm{A}$$

EXAMPLE 7.3

A numerical example may help to illustrate the geometry involved in Stokes' theorem. Consider the portion of a sphere shown in Figure 7.17. The surface is specified by $r = 4$, $0 \leq \theta \leq 0.1\pi$, $0 \leq \phi \leq 0.3\pi$, and the closed path forming its perimeter is composed of three circular arcs. We are given the field $\mathbf{H} = 6r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$ and are asked to evaluate each side of Stokes' theorem.

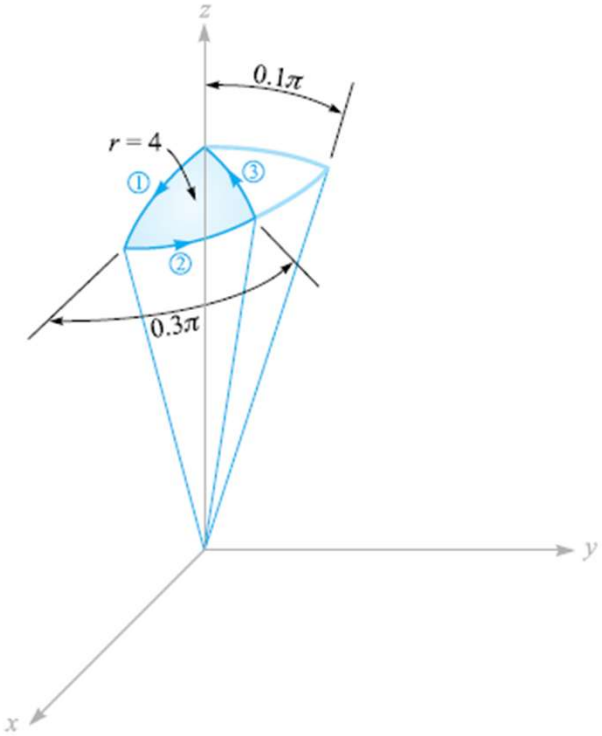


Figure 7.17 A portion of a spherical cap is used as a surface and a closed path to illustrate Stokes' theorem.

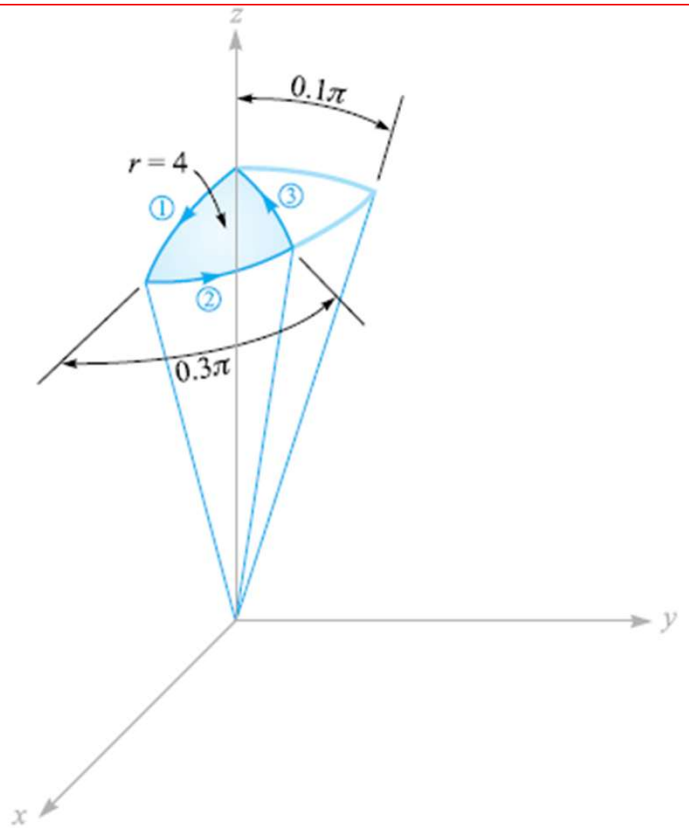


Figure 7.17 A portion of a spherical cap is used as a surface and a closed path to illustrate Stokes' theorem.

The first term is zero on all three segments of the path since $r = 4$ and $dr = 0$, the second is zero on segment 2 as θ is constant, and the third term is zero on both segments 1 and 3. Thus,

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_1 H_\theta r \, d\theta + \int_2 H_\phi r \sin\theta \, d\phi + \int_3 H_\theta r \, d\theta$$

Because $H_\theta = 0$, we have only the second integral to evaluate,

$$\oint \mathbf{H} \cdot d\mathbf{L} = \int_0^{0.3\pi} [18(4) \sin 0.1\pi \cos\phi] 4 \sin 0.1\pi \, d\phi$$
$$= 288 \sin^2 0.1\pi \sin 0.3\pi = 22.2 \text{ A}$$

We next attack the surface integral. First, we use (26) to find

$$\nabla \times \mathbf{H} = \frac{1}{r \sin\theta}(36r \sin\theta \cos\theta \cos\phi)\mathbf{a}_r + \frac{1}{r}\left(\frac{1}{\sin\theta}6r \cos\phi - 36r \sin\theta \cos\phi\right)\mathbf{a}_\theta$$

Because $d\mathbf{S} = r^2 \sin\theta \, d\theta \, d\phi \, \mathbf{a}_r$, the integral is

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \int_0^{0.3\pi} \int_0^{0.1\pi} (36 \cos\theta \cos\phi) 16 \sin\theta \, d\theta \, d\phi$$
$$= \int_0^{0.3\pi} 576 \left(\tfrac{1}{2} \sin^2\theta\right) \Big|_0^{0.1\pi} \cos\phi \, d\phi$$
$$= 288 \sin^2 0.1\pi \sin 0.3\pi = 22.2 \text{ A}$$

D7.6. Evaluate both sides of Stokes' theorem for the field $\mathbf{H} = 6xy\mathbf{a}_x - 3y^2\mathbf{a}_y$ A/m and the rectangular path around the region, $2 \leq x \leq 5, -1 \leq y \leq 1, z = 0$. Let the positive direction of $d\mathbf{S}$ be $\mathbf{a}_z$.

Ans. −126 A; −126 A